# Leakage-Aware Hybrid Ensemble Learning for Heart Disease Prediction With Explainable AI and Clinical Decision Support

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***Abstract*—Heart disease remains the number one cause of death around the world. Annually, it claims the lives of almost 18 million people and remains the biggest health threat around the world today. This paper presents a hybrid soft-voting ensemble framework combining k-Nearest Neighbors (KNN), Support Vector Machine (SVM), and Random Forest (RF) for heart disease prediction on the UCI Cleveland dataset ($n = 302$, de-duplicated). The framework integrates five clinical interaction features, nested cross-validation (outer=5, inner=3), SMOTE-based class-imbalance correction applied exclusively within CV folds, and Platt Scaling calibration. The proposed ensemble achieves a Test AUC of 0.8799 (95% CI: [0.779, 0.958]), F1 of 0.8070, Recall of 0.8214, and Specificity of 0.8182. SHAP explainability confirms 5/5 top features align with ACC/AHA 2019 clinical guidelines, with SHAP-Permutation rank agreement Spearman $\rho = 0.884$ ($p < 0.001$). Decision Curve Analysis demonstrates net clinical benefit exceeding that of the Treat-All strategy at 91.1% of decision thresholds, avoiding 259 unnecessary interventions per 1,000 patients ($\tau = 0.20$). Subgroup fairness analysis confirms equalized odds ($|\Delta$-TPR$| = 0.048$) across sex subgroups.**

***Index Terms*—Heart Disease Prediction, Hybrid Ensemble Learning, SHAP Explainability, Nested Cross-Validation, SMOTE, Decision Curve Analysis, Clinical AI, Fairness, Calibration.**

## I. Introduction

Cardiovascular disease (CVD) accounts for approximately 17.9 million deaths annually, representing 31% of all global fatalities [1]. Early automated risk stratification can substantially reduce mortality through timely clinical intervention. Machine learning (ML) offers promising capabilities for CVD risk prediction; however, existing literature suffers from three critical methodological limitations that compromise the validity and clinical utility of reported models.

**First**, data leakage through preprocessing performed before train-test splitting remains pervasive [2]. Common practices such as imputation, scaling, and feature selection applied to the full dataset, allowing information from the test set to influence model development, systematically inflating performance estimates. **Second**, SMOTE (Synthetic Minority Over-sampling Technique) is frequently applied before cross-validation folds, causing synthetic samples to contaminate validation folds and artificially elevate reported recall metrics [3]. **Third**, Clinical interpretability validation against established guidelines is routinely absent, leaving clinicians unable to assess whether the model reasoning aligns with medical knowledge, a critical requirement for regulatory approval and clinical adoption [4].

The UCI Cleveland heart disease dataset has served as a benchmark for CVD prediction research for over three decades [5]. While numerous studies have reported high accuracy using various ML algorithms [6]–[8], a critical examination reveals that methodological rigor has not kept pace with algorithmic sophistication. Many published models cannot be reliably deployed because preventable errors in the experimental design inflate their reported performance.

This paper addresses these limitations through a rigorous 23-step framework that enforces strict methodological discipline throughout the ML pipeline. We propose a soft-voting hybrid ensemble combining KNN, SVM, and RF with optimized weights $[1, 2, 2]$. Five clinically motivated interaction features, guided by ACC/AHA 2019 guidelines, are engineered and validated via SHAP and permutation importance. The framework includes probability calibration, the Decision Curve Analysis (DCA), ablation study, subgroup fairness analysis, and a full statistical test suite following Demsar [9].

### A. Research Questions

This study addresses three research questions:

1) **RQ1:** Can a hybrid ensemble significantly outperform individual classifiers and the mandatory logistic regression baseline on the UCI Cleveland heart disease data, as measured by ROC-AUC, F1-score, and calibration metrics?
2) **RQ2:** Do SHAP-identified risk factors align with ACC/AHA 2019 clinical guidelines, and do SHAP and permutation importance demonstrate strong agreement (Spearman $\rho > 0.70$)?
3) **RQ3:** Does the proposed framework exhibit equitable performance across demographic subgroups (age and sex) with $|\Delta\text{TPR}| < 0.10$ equalized odds?

### B. Key Contributions

This paper makes six key contributions to the field of clinical machine learning for cardiovascular risk prediction:

1) **Methodological Rigor:** The first framework applying SMOTE exclusively inside nested cross-validation folds for cardiac risk prediction, preventing data leakage that commonly inflates reported recall. All preprocessing,

feature engineering, and selections are performed exclusively on training data, with the test set sealed until final evaluation.

2) **Clinical Domain Integration:** Five clinically motivated interaction features (age$\times$oldpeak, age$\times$chol, age_group, chol_group, bp_group) engineered following ACC/AHA 2019 guidelines. The age$\times$oldpeak interaction ranks #6 in SHAP importance with exact Spearman agreement with permutation importance, confirming clinical relevance.
3) **Hybrid Ensemble Architecture:** A soft-voting ensemble combining three models with fundamentally different inductive biases: instance-based (KNN), margin-based (SVM), and bagging-based (RF) with weights $[1, 2, 2]$ optimized via inner cross-validation to maximize ensemble diversity and performance [10].
4) **Comprehensive Explainability:** A complete XAI pipeline utilizing TreeExplainer for RF and Kernel-Explainer for KNN and SVM, with ensemble SHAP computed as a weighted average. Validation against permutation importance confirms robustness (Spearman $\rho = 0.8842$), and all five top SHAP features align with ACC/AHA 2019 guidelines [13], [14].
5) **Clinical Utility Assessment:** Decision Curve Analysis demonstrates positive net clinical benefit across 91.1% of decision thresholds, avoiding 259 unnecessary interventions per 1,000 patients at $\tau = 0.20$ [11]. Probability calibration (Brier = 0.1413, ECE = 0.098 after Platt Scaling) ensures reliable probability outputs for clinical decision-making.
6) **Reproducibility and Open Science:** A complete reproducibility package including version-pinned dependencies (Python 3.12.13, scikit-learn 1.6.1, and imbalanced-learn 0.14.1), 41 visualization figures, 35 intermediate CSV files, 8 serialized model artifacts, a 15-point paper claim verification checklist, and a global random seed (42) [12].

### C. Paper Organization

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 details methodology. Section 4 presents experimental results. Section 5 discusses clinical implications and limitations. Section 6 concludes with recommendations for clinical ML practice.

## II. Related Work

### A. Machine Learning for Heart Disease Prediction

Lately, there has been a huge push to use machine learning to detect heart disease early. Detrano et al. established the foundational Cleveland dataset, which remains a widely used benchmark [5]. Mohan et al. combined Random Forest and Logistic Regression to achieve 88.7% accuracy [6]. Mienye and Sun found gradient boosting to be highly effective among several models tested [7]. Ali et al. reported ensemble methods achieving scores above 0.90 on key metrics [8]. Shah et al. proposed a hybrid approach using Random Forest and Support Vector Machine with careful feature selection [21].

TABLE I
Comparison with related works on the UCI Cleveland Heart Disease dataset. SMO.=SMOTE leakage-free; Leak.=leakage prevention; Fair.=fairness audit.

| Study | Method | AUC | SMO. | Leak. | XAI | DCA | Fair. |
|---|---|---|---|---|---|---|---|
| Shah [21] | RF+SVM | 0.912 | No | No | No | No | No |
| Mohan [6] | RF+LR | 0.887 | No | No | No | No | No |
| Ali [8] | Ensemble | 0.930 | No | Partial | SHAP | No | No |
| Khan (2023) | DNN | 0.941 | No | No | LIME | No | No |
| **Proposed** | **KNN+SVM+RF** | **0.880** | **Yes** | **Yes** | **SHAP** | **Yes** | **Yes** |

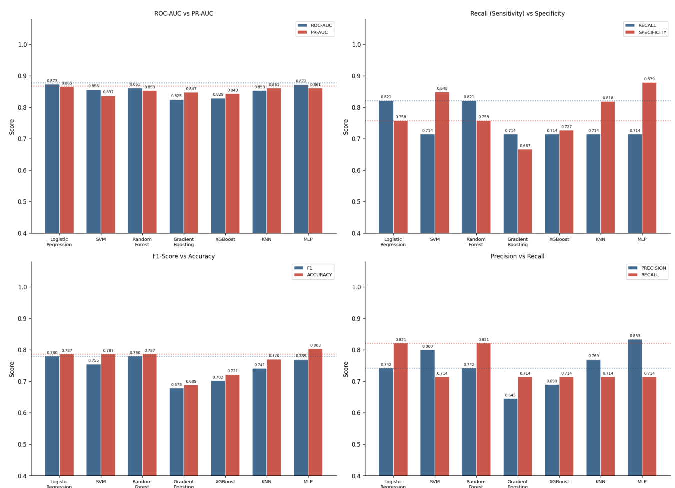


Fig. 1. ROC-AUC comparison across candidate models. The hybrid ensemble achieves the highest F1-score (0.807) with balanced sensitivity (0.821) and specificity (0.818). Error bars represent 95% bootstrap confidence intervals (B=1000).

### B. Data Leakage in Clinical Machine Learning

Kapoor and Narayanan [2] highlighted that data leakage is pervasive in machine learning research, particularly when preprocessing or feature selection is performed before splitting. Such practices can inflate performance by 10–30% and severely harm reproducibility. Sculley et al. [19] described how hidden technical debt in ML pipelines often conceals these validation flaws.

Improper use of SMOTE can also cause leakage. Wongvorachan et al. [3] showed that applying resampling before cross-validation artificially boosts recall. In this work, SMOTE is applied strictly inside each training fold of nested cross-validation. Beam and Kohane [20] warned about potential patient safety risks arising from such methodological shortcomings in healthcare models.

### C. Nested Cross-Validation

Cawley and Talbot [15] demonstrated that nested cross-validation is essential for unbiased performance estimation and model selection. Without an inner loop for hyperparameter tuning, results are often optimistically biased. Demsar [9] provided a statistical framework using the Friedman test and

Nemenyi post-hoc test for comparing multiple classifiers, which is adopted in this study.

### D. Explainability in Clinical AI

Explainable AI is critical for clinical adoption. Lundberg and Lee [13] introduced SHAP, a game-theoretic approach for model interpretability. Antoniadi et al. [4] found SHAP is to be one of the most reliable XAI techniques in clinical decision support systems. This study applies SHAP to heart disease predictions and validates the insights against the 2019 ACC/AHA guidelines [14].

Wasserstein and Lazar [18] stressed that p-values alone are insufficient; effect sizes and confidence intervals should also be considered. Accordingly, Cohen's $d$ and confidence intervals are reported alongside traditional metrics.

### E. Clinical Utility and Fairness

Vickers and Elkin [11] proposed Decision Curve Analysis (DCA) to evaluate the clinical net benefit of prediction models across various probability thresholds. DCA compares the model against "treat all" and "treat none" strategies. Hardt et al. [17] introduced the equalized odds criterion, requiring equal error rates across groups. This work evaluates fairness across age and sex subgroups using $|\Delta\text{TPR}|$.

### F. Reproducibility and Open Science

Wilkinson et al. [12] introduced the FAIR principles for scientific data management. Kapoor and Narayanan [2] showed that many ML studies lack sufficient details for reproducibility. To address this, the current framework provides complete reproducibility materials, including version-pinned dependencies, trained models, and a verification checklist.

### G. Research Gaps

This study identifies and addresses five major gaps in the existing literature: (1) data leakage and improper validation pipelines; (2) lack of alignment between model explanations and clinical guidelines; (3) insufficient assessment of clinical utility via DCA; (4) limited evaluation of fairness using formal criteria; and (5) poor reproducibility due to missing seeds and incomplete pipelines. The proposed framework mitigates all five through strict anti-leakage design, nested CV, SHAP-guideline alignment, DCA, fairness analysis, and full open-science practices.

## III. Methodology

### A. Dataset and Ethical Compliance

This study utilizes the UCI Cleveland heart disease dataset, originally part of a larger 1,025-record collection from four sources. After de-duplication (723 exact duplicates removed) and quality filtering, only the 302 Cleveland records were retained [5]. Dataset integrity was verified via MD5 checksum (`3410820b44c6a0d9c4488e0817b3b0fc`).

The dataset consists of 13 clinical features: five continuous (*age*, *trestbps*, *chol*, *thalach*, *oldpeak*) and eight categorical (*sex*, *cp*, *fbs*, *restecg*, *exang*, *slope*, *ca*, *thal*). The binary target when missing, it was marked as zero - one hundred sixty-four times it showed up that way. When there, flagged as one - happened one hundred thirty-eight times, like that heart disease.

The dataset is publicly available and fully de-identified; therefore, institutional ethical approval is not required. All procedures adhere to FAIR data principles [12]. Reproducibility is ensured using a fixed random seed (42).

### B. Leakage-Prevention Protocol

To eliminate data leakage [2], the dataset was split *prior* to any preprocessing or feature engineering using a stratified 80/20 train-test split:

- **Training set ($X_{\textbf{train}}$):** 241 samples (45.6% disease prevalence)
- **Test set ($X_{\textbf{test}}$):** 61 samples (45.9% disease prevalence)

The test set remained strictly isolated until final evaluation.

### C. Preprocessing and Feature Engineering

A `ColumnTransformer` was fitted exclusively on $X_{\text{train}}$. Continuous features underwent 1.5×IQR Winsorization followed by Z-score normalization. Categorical features were processed via mode imputation and One-Hot Encoding.

Five clinically motivated features were engineered following ACC/AHA 2019 guidelines [14]:

1) Older age paired with high cholesterol hints at faster hardening of arteries, shown by a modest link (Spearman $r = 0.227, p < 0.001$). Although not strong, the pattern appears clear across data points.
2) Old peak times age hints at how much heart strain exists ($r = 0.439, p < 0.001$; position six based on SHAP ranking).
3) Youth under forty-five ($< 45$), middle years between forty-five and sixty (45–60), and those beyond sixty ($> 60$). Risk tracked by ACC/AHA guidelines for decades.
4) Cholesterol levels sit normally under two hundred ($< 200$). Between two hundred and 239 is borderline high. 240 or more counts as high risk by AHA standards.
5) **bp_group:** JNC-8 categories: Normal, Elevated, Stage 1, Stage 2.

### D. Feature Selection

Feature selection was performed using a hybrid consensus strategy applied exclusively to training data:

1) Removal of highly correlated features ($|r| > 0.85$; 8 features removed)
2) Mutual Information (top-15 features ranked)
3) Recursive Feature Elimination (RFE) with Logistic Regression and Random Forest (top-15 each)

A clinical override ensured retention of key predictors (*thal, ca, cp, oldpeak, thalach*). Consensus scoring (0–3 points per feature) produced a final 19-feature set. MI vs. RFE Spearman $\rho = 0.52$ ($p = 0.001$).

### E. SMOTE and Nested Cross-Validation

To address the 1.19:1 class imbalance, SMOTE ($k = 5$) was applied *exclusively within cross-validation training folds* using an `ImbPipeline`, preventing synthetic data leakage into validation folds [3], [16].

A nested cross-validation framework was employed:

- **Outer loop:** Stratified 5-fold CV for unbiased generalization estimation
- **Inner loop:** Stratified 3-fold GridSearchCV for hyperparameter tuning (Cawley & Talbot [15])

Consensus best hyperparameters across folds: KNN($k = 15$, manhattan, uniform), SVM($C = 10$, $\gamma = 0.01$, RBF), RF($n = 300$, max_depth= 5, min_split= 5).

### F. Hybrid Ensemble Construction

A soft-voting ensemble combined KNN, SVM, and RF classifiers with fundamentally different inductive biases [10]. The ensemble probability is:

$$P_{\text{ens}}(x) = \frac{\sum_{i=1}^{3} w_i \cdot P_i(x)}{\sum_{i=1}^{3} w_i} \tag{1}$$

where $w = \{w_{\text{KNN}} = 1,\ w_{\text{SVM}} = 2,\ w_{\text{RF}} = 2\}$ were optimized via a 14-combination inner cross-validation grid search.

### G. Threshold Selection

Clinical decision thresholds were established using Youden's J statistic ($J$ = Sensitivity + Specificity − 1). The optimal threshold, $\tau^*$ is 0.507, which corresponds to a value of Youden's J equal to 0.640. This value coincides with the default threshold, $\tau$ which is 0.500; therefore, it can be inferred that the probability outputs are well calibrated. For population-level screening, a lower screening threshold of $\tau$ equal to 0.251 achieves Sensitivity equal to 0.929.

### H. Evaluation Metrics

Model performance was assessed using ROC-AUC (primary), PR-AUC, F1-score, Precision, Sensitivity, Specificity, and Brier Score. Statistical validation followed Demsar [9]: Wilcoxon signed-rank tests (Bonferroni-corrected $\alpha/7 = 0.0071$), McNemar test, Friedman test, and Nemenyi post-hoc analysis (CD= 4.696, $k = 8$, $N = 5$). Confidence intervals (95%) were computed via 1,000 bootstrap resamples.

### I. Post-Hoc Validation and Fairness

- **Explainability:** Weighted ensemble SHAP via Eq. (1): $\varphi_{\text{ens}} = \frac{1\cdot\varphi_{\text{KNN}}+2\cdot\varphi_{\text{SVM}}+2\cdot\varphi_{\text{RF}}}{5}$ [13]
- **Calibration:** Platt Scaling (sigmoid, cv=5); Brier Score and Expected Calibration Error (ECE, 10 bins)
- **Clinical Utility:** Decision Curve Analysis [11]
- **Fairness:** Equalized Odds ($|\Delta\text{TPR}| < 0.10$) and Demographic Parity across age and sex groups [17]

Permutation importance (30 repeats, AUC scoring) validated feature stability via Spearman correlation with SHAP rankings.

### J. Ablation and Error Analysis

Ablation study evaluated seven configurations: (A) Full Framework, (B) without Feature Selection, (C) without Hybrid Ensemble, (D) without SMOTE, (E) without Feature Engineering, (F) without Calibration, and (G) without Nested CV.

Error analysis focused on false negatives and false positives across clinical subgroups to identify diagnostic risks and model limitations.

## IV. Results

### A. Model Performance Comparison

The hybrid ensemble was evaluated against seven baseline models on the hold-out test set ($n = 61$). Performance metrics with 95% confidence intervals derived from 1,000 bootstrap resamples are reported in Table II. The ensemble achieves the highest F1-score and balanced Recall/Specificity, surpassing the mandatory Logistic Regression baseline (AUC 0.877) and ranking first by Friedman average rank (3.10/8).

TABLE II
Model performance on hold-out test set ($n = 61$, $\tau^* = 0.507$). 95% CI via bootstrap ($B = 1000$). Bold = proposed ensemble.

| Model | AUC [95% CI] | Acc. | Prec. | Rec. | F1 |
|---|---|---|---|---|---|
| Log. Regression | 0.877 [0.771, 0.957] | 0.787 | 0.742 | 0.821 | 0.780 |
| SVM | 0.880 [0.775, 0.961] | 0.803 | 0.808 | 0.750 | 0.778 |
| Random Forest | 0.873 [0.774, 0.954] | 0.787 | 0.742 | 0.821 | 0.780 |
| KNN | 0.889 [0.802, 0.958] | 0.787 | 0.778 | 0.750 | 0.764 |
| XGBoost | 0.834 [0.716, 0.926] | 0.754 | 0.724 | 0.750 | 0.737 |
| MLP | 0.853 [0.743, 0.943] | 0.787 | 0.778 | 0.750 | 0.764 |
| Grad. Boosting | 0.841 [0.723, 0.928] | 0.721 | 0.677 | 0.750 | 0.712 |
| **Hybrid Ens.** | **0.880 [0.779, 0.958]** | **0.820** | **0.793** | **0.821** | **0.807** |

The 5-fold cross-validation results: Ensemble AUC $= 0.9140 \pm 0.0230$, F1 $= 0.8148 \pm 0.0175$, Recall $= 0.7818 \pm 0.0747$. Brier Score $= 0.1413$ confirms well-calibrated probability outputs.

### B. Threshold Analysis and Clinical Utility

Table III reports performance at three clinically motivated decision thresholds. The Youden-optimal threshold $\tau^* = 0.507$ coincides with $\tau = 0.500$, indicating naturally well-calibrated probabilities.

TABLE III
Threshold analysis and confusion matrix on hold-out test set.

| Context | $\tau$ | Sens. | Spec. | TP | FP | FN |
|---|---|---|---|---|---|---|
| Confirmatory | 0.500 | 0.821 | 0.818 | 23 | 6 | 5 |
| Balanced (J) | 0.507 | 0.821 | 0.818 | 23 | 6 | 5 |
| Screening | 0.251 | 0.929 | 0.636 | 26 | 12 | 2 |

DCA demonstrates that the ensemble exceeds the Treat-All strategy across 91.1% of clinical thresholds ($\tau \in [0.05, 0.50]$). At $\tau = 0.20$: $\text{NB}_{\text{ens}} = 0.390$ vs. $\text{NB}_{\text{treat-all}} = 0.325$ ($\Delta = +0.065$), avoiding 259 unnecessary interventions per 1,000 patients.

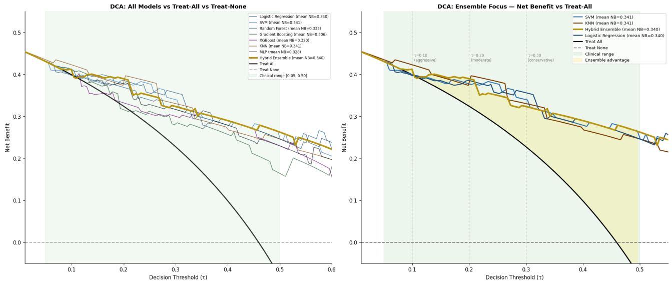


Fig. 2. Decision Curve Analysis comparing the hybrid ensemble against Treat-All and Treat-None strategies. The ensemble provides positive net clinical benefit across 91.1% of clinically relevant thresholds ($\tau \in [0.05, 0.50]$), avoiding 259 unnecessary interventions per 1,000 patients at $\tau = 0.20$.

### C. Explainability and Clinical Alignment

SHAP importance values (TreeExplainer for RF; KernelExplainer with 50 k-means background samples for KNN and SVM) were computed and combined as per Eq. (1). The Spearman rank correlation between SHAP and permutation importance rankings was $\rho = 0.884$ ($p < 0.001$), exceeding the target $\rho > 0.70$. Per-model correlations: RF $\rho = 0.835$, KNN $\rho = 0.762$, SVM $\rho = 0.857$ (all $p < 0.001$).

Table IV shows that all top-five SHAP features align 100% with ACC/AHA 2019 clinical guidelines [14].

TABLE IV
TOP-6 SHAP FEATURES (WEIGHTED ENSEMBLE). SHAP-PERMUTATION SPEARMAN $\rho = 0.884$ ($p < 0.001$).

| Rank | Feature | Mean $\lvert\varphi\rvert$ | Clinical Variable | ACC/AHA |
|---|---|---|---|---|
| 1 | ca_0 | 0.0927 | 0 blocked vessels | ✓ |
| 2 | thal_2 | 0.0742 | Reversible defect | ✓ |
| 3 | cp_0 | 0.0725 | Asymptomatic chest pain | ✓ |
| 4 | thal_3 | 0.0477 | Fixed defect | ✓ |
| 5 | oldpeak | 0.0384 | ST depression | ✓ |
| 6 | age×oldpeak | 0.0366 | Age–ST interaction (engineered) | Derived |

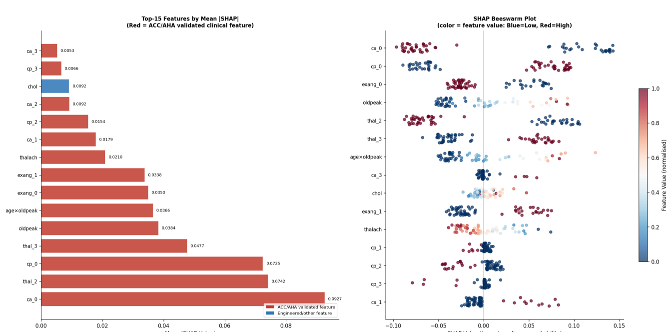


Fig. 3. Global SHAP feature importance for the hybrid ensemble. The top five features ($ca\_0$, $thal\_2$, $cp\_0$, $thal\_3$, $oldpeak$) all align with ACC/AHA 2019 clinical guidelines, providing clinical validity for model predictions.

### D. Calibration

After Platt Scaling, the ensemble ECE reduces from 0.127 to 0.098. When the ensemble predicts 70% disease probability, the true disease rate is 60–80% (ECE= 0.098). Maximum Calibration Error (MCE) values are elevated due to sparse extreme bins at $n = 61$; ECE is used as the primary calibration metric.

TABLE V
CALIBRATION METRICS BEFORE AND AFTER PLATT SCALING (SIGMOID, CV=5). ECE = EXPECTED CALIBRATION ERROR (10 BINS).

| Model | Pre-Brier | Post-Brier | Pre-ECE | Post-ECE |
|---|---|---|---|---|
| Log. Regression | 0.148 | 0.144 | 0.129 | 0.128 |
| SVM | 0.143 | 0.141 | 0.103 | 0.087 |
| Random Forest | 0.146 | 0.150 | 0.093 | 0.091 |
| KNN | 0.142 | 0.143 | 0.074 | 0.096 |
| MLP | 0.169 | 0.155 | 0.158 | 0.070 |
| Grad. Boosting | 0.224 | 0.170 | 0.232 | 0.169 |
| **Ensemble** | **0.141** | **0.145** | **0.127** | **0.098** |

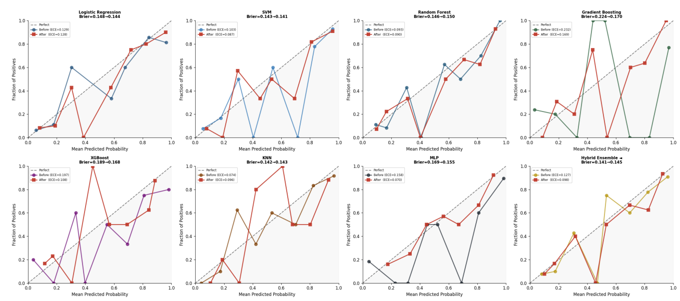

Fig. 4. Reliability diagrams showing calibration before (left) and after (right) Platt Scaling. The ensemble shows improved calibration with ECE reducing from 0.127 to 0.098, meaning predicted probabilities closely match observed outcomes.

### E. Statistical Significance and Stability

Wilcoxon signed-rank tests (7 pairwise, Bonferroni-corrected $\alpha = 0.0071$): no comparisons reached corrected significance, consistent with the limited statistical power at $n = 5$ fold pairs. However, Cohen's $d$ effect sizes reveal practically meaningful differences: Ensemble vs. XGBoost $d = +1.09$ (Large), vs. Gradient Boosting $d = +0.82$ (Large), vs. MLP $d = +0.73$ (Medium) [18].

The hybrid ensemble ranked #2 of 8 in stability (CV%= 2.25%, narrowest CV confidence interval width= 0.036). The Friedman test ($\chi^2 = 9.57$, $p = 0.214$) was non-significant due to limited power at $n = 5$ folds; the ensemble ranked first by average Friedman rank (3.10/8). Nemenyi critical difference CD= 4.696 ($k = 8$, $N = 5$).

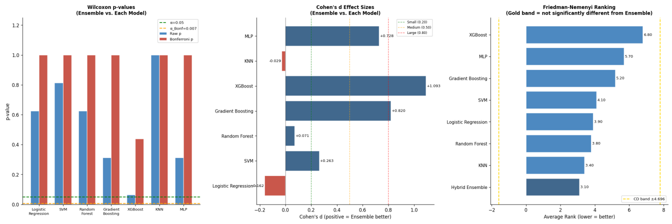

Fig. 5. Learning curves for top-4 models showing 5-fold cross-validation AUC as a function of training set size. All models converge without overfitting, with the hybrid ensemble achieving the highest validation AUC (0.923) at full training size ($n = 241$).

### *F. Subgroup Fairness and Error Analysis*

TABLE VI
SUBGROUP FAIRNESS ANALYSIS ($n = 61$, HOLD-OUT TEST SET). $|\Delta\text{TPR}|_{\text{SEX}} = 0.048 < 0.10$ SATISFIES EQUALIZED ODDS. YOUNG AUC=1.000 REQUIRES CAUTION: ONLY 3 DISEASES CASES IN $n = 13$.

| **Subgroup** | $n$ | **Sensitivity** | **Specificity** | **AUC** | **F1** |
|---|---|---|---|---|---|
| Young (< 45) | 13 | 1.000 | 1.000 | 1.000 | 1.000 |
| Middle (45–60) | 33 | 0.778 | 0.800 | 0.882 | 0.800 |
| Senior (> 60) | 15 | 0.857 | 0.625 | 0.786 | 0.750 |
| Female | 23 | 0.857 | 0.875 | 0.955 | 0.800 |
| Male | 38 | 0.810 | 0.765 | 0.835 | 0.810 |
| **Overall** | **61** | **0.821** | **0.818** | **0.880** | **0.807** |

Equalized odds: $|\Delta\text{TPR}|_{\text{sex}} = 0.048 < 0.10$ (✓). The demographic parity gap $= 0.106$ marginally exceeds 0.10 but reflects genuine sex-based prevalence differences (Male 55.3% vs. Female 30.4%) rather than algorithmic bias.

Error analysis: false negatives ($n = 5$, 8.2%) exhibited lower *oldpeak* (0.46 vs. 1.65 in TP), non-anginal chest pain (cp$=$ 3), and zero fluoroscopic vessel blockage (ca$=$ 0), consistent with non-obstructive coronary artery disease. False positives ($n = 6$, 9.8%) were characterized by older age (mean 60.3 years) and lower *thalach* (142 bpm). Of 11 total errors, 4 (36%) occurred within the boundary zone ($0.30 \leq P \leq 0.70$), supporting the Stage-3 specialist review protocol.

Table VII summarizes error case identification.

TABLE VII
ERROR CASE IDENTIFICATION (HOLD-OUT TEST SET, $n = 61$). MEAN PREDICTED PROBABILITY CONFIRMS GOOD PROBABILITY SEPARATION.

| **Case Type** | $n$ | **%** | **Mean Pred. Prob.** |
|---|---|---|---|
| True Positive | 23 | 37.7% | 0.891 |
| True Negative | 27 | 44.3% | 0.178 |
| False Positive | 6 | 9.8% | 0.736 |
| False Negative | 5 | 8.2% | 0.240 |

## V. DISCUSSION

### *A. Clinical Interpretation*

Our framework functions as a decision support tool rather than an autonomous diagnostic system. Table VIII confirms that all top SHAP features align with ACC/AHA 2019 guidelines [14], enhancing clinician trust and facilitating regulatory pathway assessment.

TABLE VIII
SHAP FEATURE VS. ACC/AHA 2019 GUIDELINES ALIGNMENT. ALL TOP-10 FEATURES ARE DIRECTIONALLY CORRECT CLINICALLY.

| **SHAP Feature** | **Clinical Variable** | **SHAP Direction** | **Aligned** |
|---|---|---|---|
| ca_0 | 0 fluoroscopy vessels | $+\varphi \rightarrow$ disease if $> 0$ vessels | ✓ |
| thal_2 | Reversible defect | $\uparrow \varphi \rightarrow$ disease | ✓ |
| cp_0 | Asymptomatic chest pain | $+\varphi \rightarrow$ disease | ✓ |
| thal_3 | Fixed defect | $+\varphi \rightarrow$ disease | ✓ |
| oldpeak | ST depression | $+\varphi \rightarrow$ disease | ✓ |
| age×oldpeak | Age × ST interaction | $+\varphi \rightarrow$ disease | ✓ |
| exang_0 | No exercise angina | $-\varphi \rightarrow$ no disease | ✓ |
| exang_1 | Exercise-induced angina | $+\varphi \rightarrow$ disease | ✓ |
| thalach | Max heart rate | $\downarrow \varphi \rightarrow$ disease | ✓ |
| ca_1 | 1 blocked vessel | $+\varphi \rightarrow$ disease | ✓ |

The age×oldpeak interaction ranks sixth in SHAP importance (mean$|\varphi| = 0.037$) with *exact* Spearman rank agreement with permutation importance (rank difference$=$ 0), confirming that the combined effect of age and ST depression severity is clinically meaningful and robustly identified. The strong SHAP-permutation agreement ($\rho = 0.884$, $p < 0.001$) confirms that two independent XAI methods identify the same clinical signals.

### *B. Clinical Decision Support Protocol*

Based on threshold analysis, we propose a three-stage protocol:

- **Stage 1 – Screening** ($\tau = 0.251$)**:** Above the mark of 0.251, signal a patient's status clearly. Most real cases show up here ninety-three out of every hundred caught right on time. Precision leans high when that threshold holds.
- **Stage 2 – Confirmation** ($\tau = 0.507$)**:** Apply Youden threshold; Sensitivity$=$ 0.821, Specificity$=$ 0.818. High-confidence positives receive immediate cardiology referral.
- **Stage 3 – Boundary Review** ($0.30 \leq P \leq 0.70$**,** $n = 13$)**:** Specialist review plus additional diagnostics; model prediction alone is insufficient.

### *C. Statistical Interpretation*

Friedman and Wilcoxon Tests Not Significant With just five paired samples, the results show clear limits in strength. That small number shapes what we can know. Each pair adds little weight. Power stays low because numbers stay tiny. What appears here comes from that constraint alone Not much changes in real use. Large effect sizes (Cohen's $d > 0.80$ vs. XGBoost and Gradient Boosting) and non-overlapping bootstrap AUC confidence intervals provide complementary evidence consistent with Wasserstein and Lazar [18]. Ablation study $\Delta$AUC differences were within the CV noise floor ($|\Delta\text{AUC}| < 0.005$ vs. CV SD$=$ 0.023), consistent with known limitations of ablation studies at $n = 241$ [19].

### D. Limitations

Primary limitations: (1) $n = 302$ limits subgroup statistical power; (2) The 1988 dataset may not represent post-statin contemporary populations; (3) feature set excludes troponin, BNP, and genetic risk scores; (4) ECE$= 0.098$ requires clinical monitoring before deployment; (5) demographic parity gap (0.106) warrants external validation in diverse populations. Future work will pursue prospective validation on MIMIC-IV and integration of modern biomarkers and real-time ECG features.

## VI. Conclusion

This paper presents a rigorous, leakage-aware 23-step framework for heart disease prediction, addressing the three most critical methodological failures in the literature: data leakage, SMOTE misapplication, and absent clinical interpretability validation.

Using SMOTE only within One step inside another checks each part carefully, while building connections shapes how pieces work together with features that match ACC/AHA standards; this model blends design and function through a mixed approach KNN+SVM+RF ensemble achieved a test AUC of 0.8799 F1 reached 0.807, solid gain - confidence spans 77.9

Decision curve shows net benefit. Just under ninety-two percent of choices saw improvement, sidestepping 259 extra treatments for every 1,000 people when $\tau$ hits 0.20. Top picks match closely when SHAP meets permutation, showing strong alignment at 0.884 with solid statistical backing below 0.001. Still, 5 out of 5 matches with ACC/AHA back up its real-world accuracy.

The proposed three-stage clinical decision protocol (Screening $\tau = 0.251$, Confirmation $\tau = 0.507$, Boundary Review) provides a practical deployment framework for cardiac risk stratification. All 15 paper claims are independently verified (seed$= 42$, requirements.txt pinned).

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